# MASTER BILL OF QUANTITIES (BOQ) & BALANCE OF SYSTEM (BOS) SCHEDULE

140.40 kWp DC / 110 kW AC Grid-Connected Rooftop Solar PV Installation
Al Noor Packaging Factory — Al Quoz, Dubai, UAE

| | | | |
|---|---|---|---|
| **Project Identifier** | AN-PV-PRJ-2026-BOQ-REV4 | **System Capacity** | 140.40 kWp DC / 110.00 kW AC |
| **Client Name** | Al Noor Packaging LLC | **Grid Connection** | DEWA 3-Phase, 400V, 50Hz |
| **Target Deployment** | Rooftop Concrete Deck (L1 Landscape) | **Export Constraint** | 100% Zero-Export Throttled |

# 1. Primary Hardware & Structural BOS Line Items

| Item | Component Category | Technical Sizing Model & Manufacturer Rating Specs | Qty | Unit |
|---|---|---|---|---|
| 1 | **Primary PV Modules** | LONGi Hi-MO X6 Explorer LR5-72HTH-585M. Tier-1 High-Efficiency HPBC Monocrystalline cells. Sized at 585W STC Nameplate, Voc=52.36V, Isc=14.15A, Vmpp=44.21V, Impp=13.24A. Physical dimensions: 2278mm x 1134mm x 35mm. Net mass: 22.0 kg. | **240** | Pcs |
| 2 | **Central Core Inverter** | Huawei SUN2000-110KTL-M2 Three-Phase Grid-Tied String Inverter. Rated AC Nameplate capacity 110kW @ 400V, 50Hz. Equipped with 10 independent MPPT channels and 20 pairs of DC input MC4 terminal blocks. Integrated smart string-level monitoring, cooling fins, and electronic load break switches. | **1** | Unit |
| 3 | **Rooftop DC Disconnect Boxes** | Rooftop External IP65 Weatherproof UV-stabilized polycarbonate combiner enclosures. Outfitted with an integrated mechanical heavy-duty multi-pole Rotary DC Load-Break Isolator Switch handle. Sized for local physical safety lockout of 6 circuits simultaneously. | **2** | Units |
| 4 | **DC Overvoltage Protection SPDs** | Phoenix Contact VAL-MS-T1/T2 1000DC-PV/2+V pluggable DIN-rail surge arrester cartridges. Rated Type 1 + Type 2 combined lightning current and surge protection. Continuous operational voltage ceiling Ucpv = 1,000V DC. Sized for parallel grounding pathways (6 units pre-nested per DC box). | **12** | Pcs |
| 5 | **Main AC Protection Switch** | ABB SACE Tmax XT3N 200A 3-Pole 400V Molded Case Circuit Breaker (MCCB) (Breaker Q). Short-circuit fault interrupting breaking capacity rated at 35kA to 50kA. Sized to satisfy the 125% continuous over-dimensioning safety multiplier above the inverter peak current (158.78A). | **1** | Unit |
| 6 | **AC Conductor Home Run** | 4-Core 3/0 AWG (85 mm² metric equivalence) Copper Conductor (Cu/XLPE/SWA/PVC) heavy-duty multi-core steel-wire armored underground/ladder power cable. Continuous run from rooftop inverter output down to main facility low-voltage distribution busbar. | **377.0** | Feet |
| 7 | **DC String Conductor Wire** | 12 AWG (4 mm²) PV1-F single-core copper photovoltaic wire line with Cross-Linked Polyethylene (XLPE) double-layer insulation jacket. Electron beam cross-linked material UV and ozone certified for continuous 1,100V DC exposure. | **2,970.8** | Feet |
| 8 | **Telemetry Data Logger** | Huawei SmartLogger 3000A Communication Gateway Platform. Functioning as the local Modbus-RTU RS485 Master controller. Coordinates high-speed communication polling loops between meters, sensors, and the inverter internal processor. | **1** | Unit |
| 9 | **Industrial Cellular Router** | Teltonika RUT955 Industrial Cellular 4G LTE Router, Gateway, and Network Switch block. Din-rail mounted with dual SIM card automatic failover, digital IO lines, and secure encrypted VPN client tunnels to deliver wireless cloud dashboard monitoring telemetry. | **1** | Unit |

| | | | | |
|---|---|---|---|---|
| 10 | **Smart Billing Net-Meter** | DEWA-Approved Three-Phase Multi-Function Smart Net-Meter / kWh Billing Energy Meter. Paired with a set of three (3) matching analog Current Transformer (CT) sensing split-core rings clamped directly downstream of the pre-existing 630A main incoming breaker. | **1** | Set |
| 11 | **Control Cabinet Power Supplies** | Industrial DIN-rail mounted Switch Mode Power Supplies (SMPS) converting single-phase 230V AC grid power into stabilized low-voltage outputs. Includes 1 x 24V DC unit (Logger core power) and 1 x 5V DC unit (Router telecom power). | **2** | Units |
| 12 | **Auxiliary Control Fuses** | 2A 250V AC Slow-Blow (Time-Delay) 10x38mm cylindrical ceramic control protection fuses. Mounted inside matching finger-safe fuse holders with active visual LED blown-fuse indicators to suppress power supply input transformer inrush current spikes. | **2** | Units |
| 13 | **Rooftop Structural Racking** | Anodized Marine-Grade Aluminum L1 Click-Fit Landscape Fixed-Tilt Racking rails and click-clamps. Pre-calculated for a fixed 10° Landscape orientation layout. Includes mid-clamps, end-clamps, and layout adapter brackets to secure panels without waterproofing punctures. | **240** | Mounts |
| 15 | **Rooftop Cable Trays** | Perforated hot-dip galvanized steel open-mesh wire baskets (50mm x 150mm profile) complete with clip-on weather shielding covers and tracking brackets to elevate DC home runs away from roof-deck stagnant rainwater traps. | **180** | Meters |
| 16 | **Equipotential Grounding System** | Structural grounding installation balance. Comprising continuous bare 6 AWG solid copper string row jumper wires, tinned-copper compression lugs, stainless steel panel frame star-washers, 16 $mm^2$ yellow/green insulated grounding core wire, and 3 x 3-meter copper-clad steel ground rods driven into the soil. | **1** | Lot |

## 2. Critical Procurement Notes & Installation Mandates

• **DC String Fuse Exemption:** Per DEWA and IEC 61215 grid codes, string fuses are legally omitted because no more than two (2) strings are paired in parallel per MPPT channel. Direct string wiring runs straight into the inverter MC4 terminations, bypassing extra fuse holders.

• **Phoenix Contact SPD Voltage Warning:** The selected VAL-MS-T1/T2 1000DC-PV surge devices carry a strict 1,000V DC maximum continuous operating voltage. In hot landscape orientation, the 20-panel string peak cold-winter open-circuit surge can mathematically spike up to 1,087.2V DC. Procurement must strictly verify that the inverter input firmware limitations or the high-headroom Phoenix Contact 1500V alternate class variation is cross-deployed to prevent micro-conduction aging loops.

• **Cable Shielding & Trays:** All 2,971 ft of DC cabling must run completely nested within the 180-meter galvanized mesh trays to isolate lines from direct desert thermal radiation and prevent mechanical abrasion against the concrete roofing membrane.

• **Grounding Path Impedance Target:** The complete lot of earth-pit copper rods and equipotential frame star-washers must yield a tested closed-loop path impedance **R ≤ 1.0 Ohm** during pre-energization field tests to pass DEWA shams inspector authorization signatures.